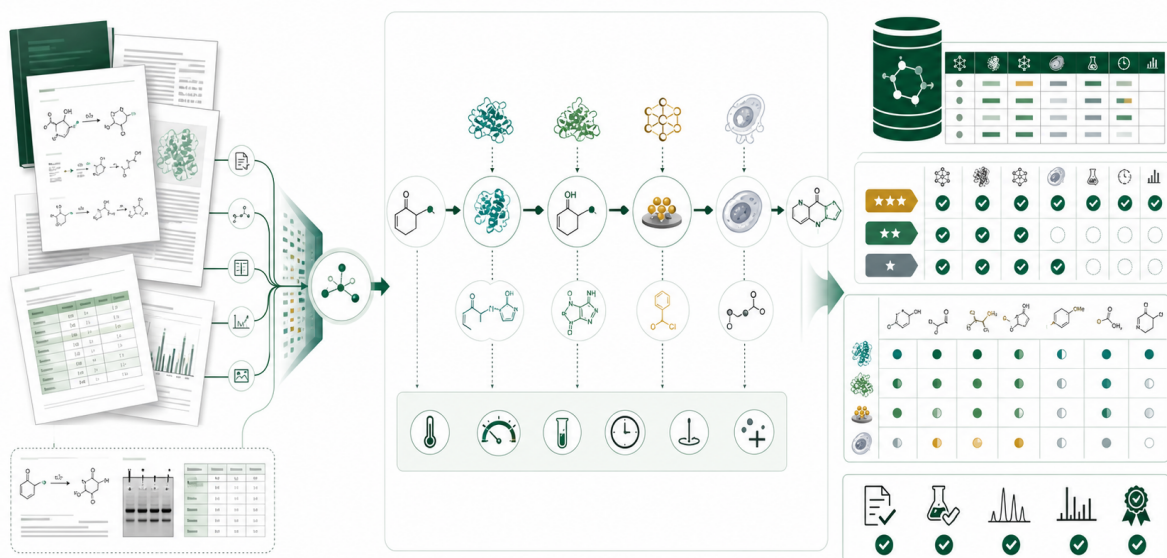


A literature-mined biocatalytic and chemoenzymatic cascade reaction database

[Author names and affiliations to be inserted before submission]

Article type: Data Descriptor

Graphical Abstract



Graphical overview of Cascade Dataset v4 construction, linking primary articles and supplementary information to evidence-aligned cascade reactions, step-level catalysts, reaction conditions, substrate-scope entries, quality tiers, and validation outputs.

Abstract

Cascade and tandem reactions are widely used to combine multiple catalytic steps into compact synthetic workflows, but machine-readable data linking step sequences, catalysts, reaction conditions, outcomes, and substrate scope remain scattered across primary articles and supplementary information. We present Cascade Dataset v4, a literature-mined database containing 3,810 cascade reaction records from 2,498 unique DOI records. The release defines a high-quality subset of 3,744 gold or silver reactions from 2,464 DOI records, including 8,609 normalized steps, 9,444 catalyst components, 21,225 input/output species records, and 3,458 substrate-scope entries. A

final supplementary-information enrichment pass added 886 rows with at least one new field value, including 246 overall yields, 173 total reaction-time values, 376 step reaction-time values, and 2,550 substrate-scope entries. The data are distributed as flat CSV, nested JSONL, and a normalized SQLite database, together with quality tiers, field coverage reports, and supplementation backlogs.

1 Background and Summary

Cascade catalysis joins multiple transformations through shared intermediates, compartmentalized or sequential operations, cofactor cycling, whole-cell systems, immobilized catalysts, organometallic partners, and biocatalytic steps. These systems are important for synthesis planning, catalyst engineering, route intensification, and biomanufacturing because they can reduce intermediate isolation while combining complementary selectivity and reactivity across enzymes and chemical catalysts.¹⁻⁴

The associated literature is difficult to reuse directly. Reported examples may emphasize overall yield, individual step conversion, stereoselectivity, catalyst compatibility, pH, temperature, reaction time, or substrate-scope tables. Step sequences and catalyst identities are often embedded in prose, reaction schemes, supplementary documents, or table footnotes. These reporting patterns limit direct use in modelling, retrosynthetic analysis, catalyst selection, and comparative studies unless the literature is converted into a structured and auditable representation.

Cascade Dataset v4 was assembled to convert this literature into a reusable, step-resolved dataset. The source database contains 3,810 reaction rows from 2,498 unique DOI records. Quality assessment assigns 2,885 rows to gold, 859 rows to silver, 60 rows to bronze, and 6 rows to quarantine. The default high-quality release is the gold plus silver subset, containing 3,744 reaction rows from 2,464 unique DOI records. Bronze and quarantine records are retained in audit files but are not recommended for default model training without manual review.

The high-quality release normalizes the nested reaction structure into 8,609 steps, 9,444 catalyst components, 21,225 species rows, and 3,458 substrate-scope entries. The dataset preserves DOI-level provenance, quality-tier labels, source evidence where available, field-coverage summaries, and supplementation backlogs so that users can select subsets matching their tolerance for uncertainty.

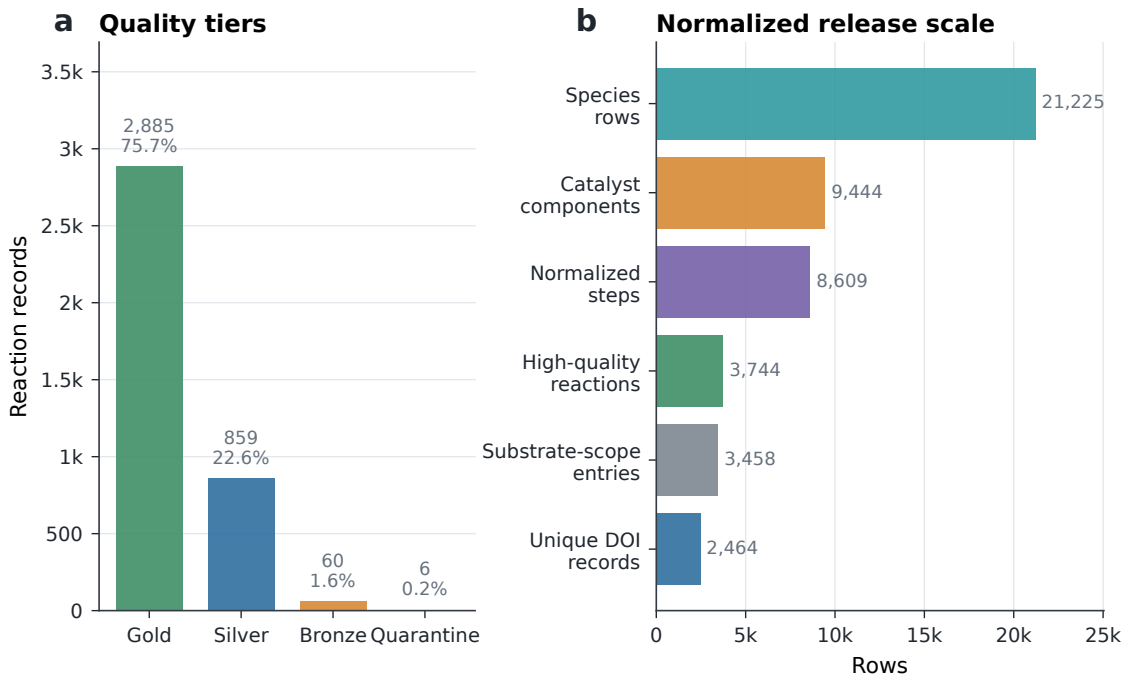


Figure 1: Cascade v4 quality tiers and normalized release scale. Counts of gold, silver, bronze, and quarantine records in the full source database, together with row counts for the high-quality release and normalized tables.

2 Methods

2.1 Literature and supplementary-information processing

The database combines primary article metadata, article-derived reaction records, supplementary-information downloads, text extraction, and LLM-assisted field completion. Primary records were organized by DOI and processed into reaction-level cascade entries. Supplementary-information processing used document-style supplementary files and skipped raw data packages or oversized data bundles. Candidate values from the enrichment pass were accepted only when the extraction confidence threshold was met and when source evidence was available.

The final audit reports 2,128 supplementary-information queue files, 2,025 unique queued DOI records, 1,788 DOI-level text caches, 1,784 usable supplementary-information text DOI records, and 1,741 LLM OK DOI records. The LLM audit recorded 2,689 cascade update candidates, including 2,112 trusted update candidates and 2,322 candidates with evidence. These metadata are retained as audit outputs rather than treated as a substitute for source provenance.

2.2 Data model

Each reaction row contains top-level cascade descriptors, target product and starting material fields, overall outcomes, global conditions, compatibility annotations, substrate-scope rows, and a step list. Each step stores a role, transformation class and name, input and output species, catalyst components, step conditions, step outcomes, and reaction SMILES where available.

The normalized SQLite export separates this structure into `reactions`, `steps`, `catalysts`, `species`, and `substrate_scope` tables. The flat CSV export keeps one row per high-quality

reaction for broad compatibility, while the JSONL export preserves nested reaction, step, condition, compatibility, and substrate-scope structures for users who need record-level context.

2.3 Quality tiers

Gold rows have strong structural identifiers, catalyst information, outcome or condition support, and no major quality flags. Silver rows are usable but retain one or more enhancement gaps, most often missing quantitative outcome, conditions, or SMILES. Bronze records are retained for retrieval and future curation. Quarantine records are structurally weak or suspicious and are excluded from the default high-quality set.

The quality assessment preserves a complete all-quality index so that users can trace why each source reaction was included in, or excluded from, the default high-quality release. This separation avoids deleting lower-confidence records while keeping the recommended modelling subset conservative.

2.4 Supplementary-information enrichment

The supplementary-information enrichment pass compared the current database with the pre-enrichment baseline. It added at least one field value to 886 rows. Top-level gains included 246 overall-yield values, 173 total reaction-time values, 74 overall-ee values, 17 overall-dr values, and 4 overall-de values. Condition gains included 140 reaction-time values, 125 temperature values, 118 pH values, 130 buffer names, and 51 solvent names. Step-level gains included 376 reaction-time values, 341 temperature values, 297 pH values, 151 solvent values, and 122 step-yield values. The pass also added 2,550 substrate-scope entries to rows that previously had no normalized substrate-scope entries.

2.5 Reproducible analysis and figure generation

The manuscript tables and figures were generated from the v4 release and quality-audit outputs using a dedicated analysis script. The script reads the release data package, writes summary CSV tables, computes release-scale and coverage statistics, and generates the manuscript figures in PNG and PDF formats. The generated figures and tables are intended to document the current release; users should analyze the released database directly for formal downstream studies.

3 Data Records

The v4 release is organized around three complementary representations: a flat high-quality CSV file, a nested JSONL file, and a normalized SQLite database. The release also includes tier-specific reaction tables, normalized table exports, an all-quality index, field-coverage audits, enrichment-increment summaries, and supplementation backlogs. The public deposition should assign a persistent DOI to the release package before journal submission. A file-level inventory is provided in Appendix A.

The primary high-quality files contain 3,744 gold or silver reactions. The normalized SQLite database contains 3,744 reaction rows, 8,609 step rows, 9,444 catalyst-component rows, 21,225 species rows, and 3,458 substrate-scope rows. The all-quality index contains all 3,810 source reaction rows with quality-tier labels, quality flags, and provenance metrics. Article metadata are available for 3,508 source rows, corresponding to 92.1% of the source reaction records.

Publisher representation in the high-quality subset is broad: Wiley accounts for 1,103 records, ACS for 797, Elsevier for 781, RSC for 478, MDPI for 119, Springer for 88, Nature for 69, and Taylor & Francis for 15, with 294 records lacking joined publisher metadata in the current article database. Publication-year metadata span 2005 through the partial 2026 record set, with 2024 and 2025 contributing the largest joined yearly counts.

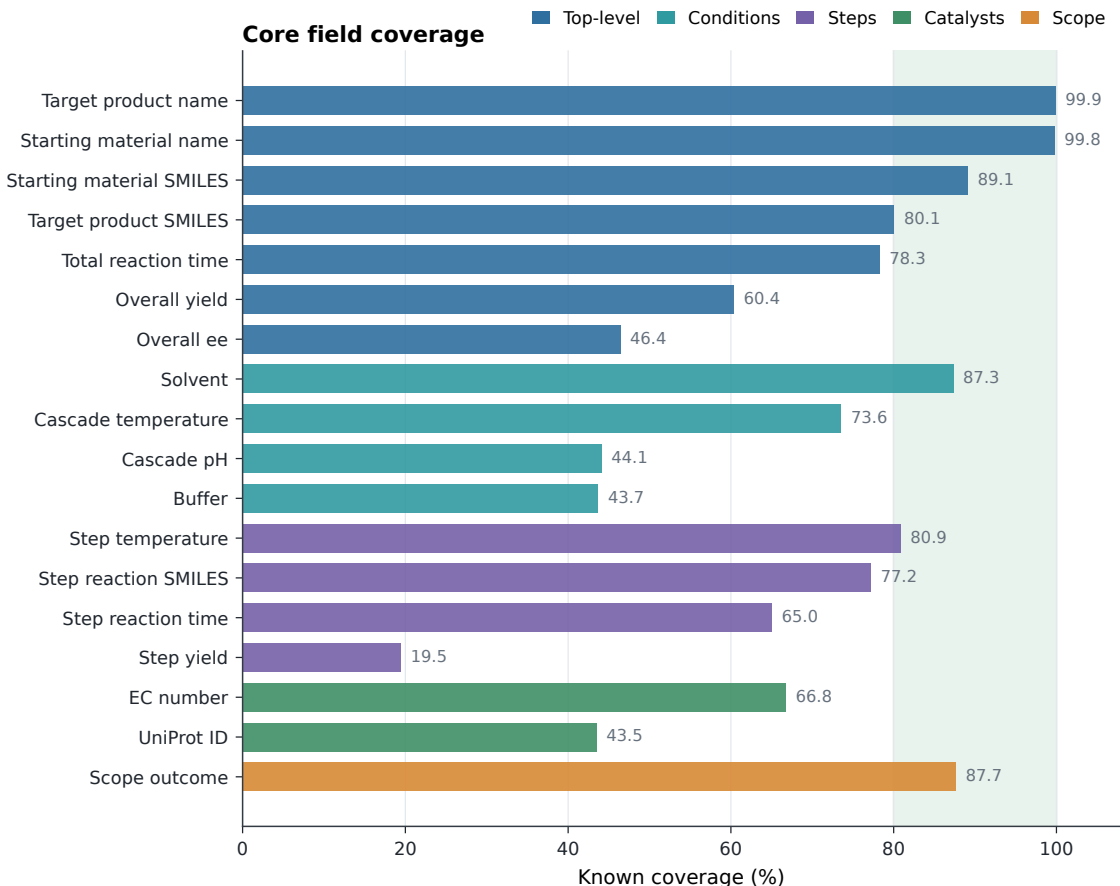


Figure 2: Core field coverage. Known-value coverage for selected top-level, condition, step, catalyst, and substrate-scope fields.

4 Data Overview

Cascade Dataset v4 captures a diverse reaction landscape. Among high-quality normalized steps, the most frequent transformation superclasses are oxidation (1,741 steps, 20.22%), reduction (1,208, 14.03%), C–C coupling (939, 10.91%), racemization (900, 10.45%), acylation (749, 8.70%), amination (664, 7.71%), and hydrolysis (642, 7.46%). Additional classes include functional-group interconversion, glycosylation, phosphorylation, isomerization, esterification, dehalogenation, amidation, and epoxide hydrolysis.

Catalyst components are dominated by enzymes but also include metal catalysts, organocatalysts, acid/base catalysts, reagent-mediated steps, whole-cell systems, photocatalysts, hybrid catalysts, nanozymes, electrocatalysts, and phase-transfer catalysts. In the normalized catalyst table, enzymes account for 5,946 components (62.96%), metal catalysts for 1,441 (15.26%), organocatalysts for 567

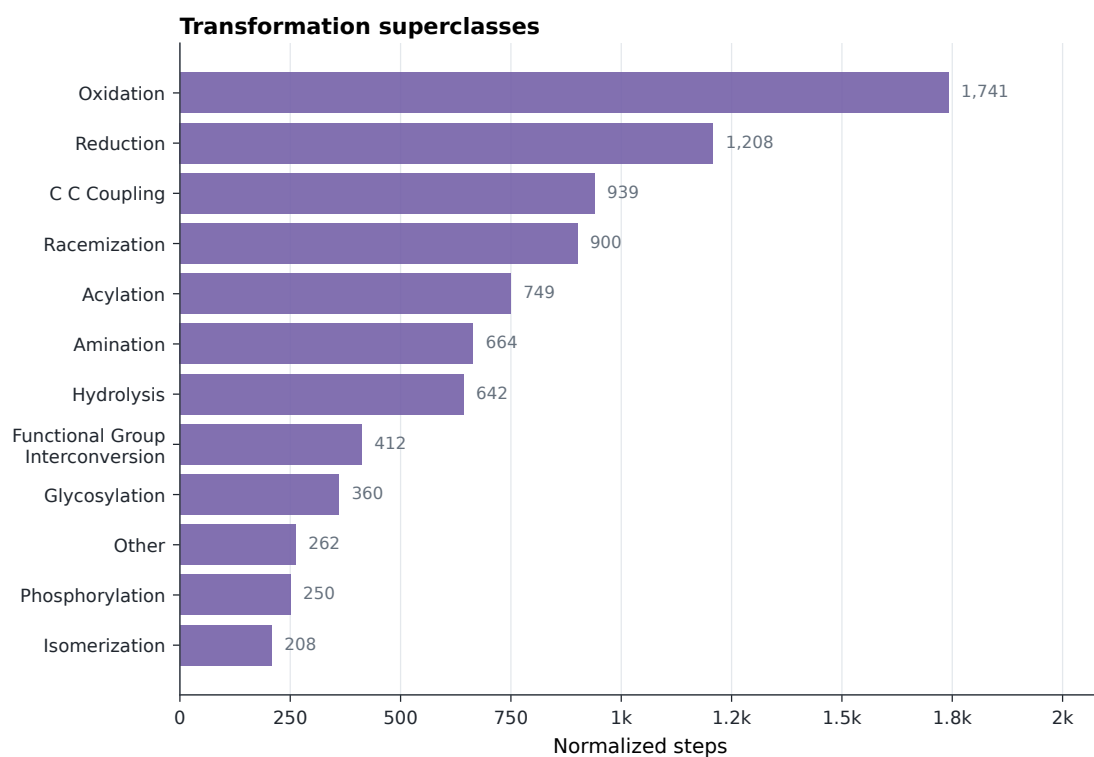


Figure 3: Common transformation superclasses. Most frequent step-level transformation classes in the high-quality subset.

(6.00%), acid/base catalysts for 453 (4.80%), reagent-mediated components for 448 (4.74%), and whole-cell components for 308 (3.26%).

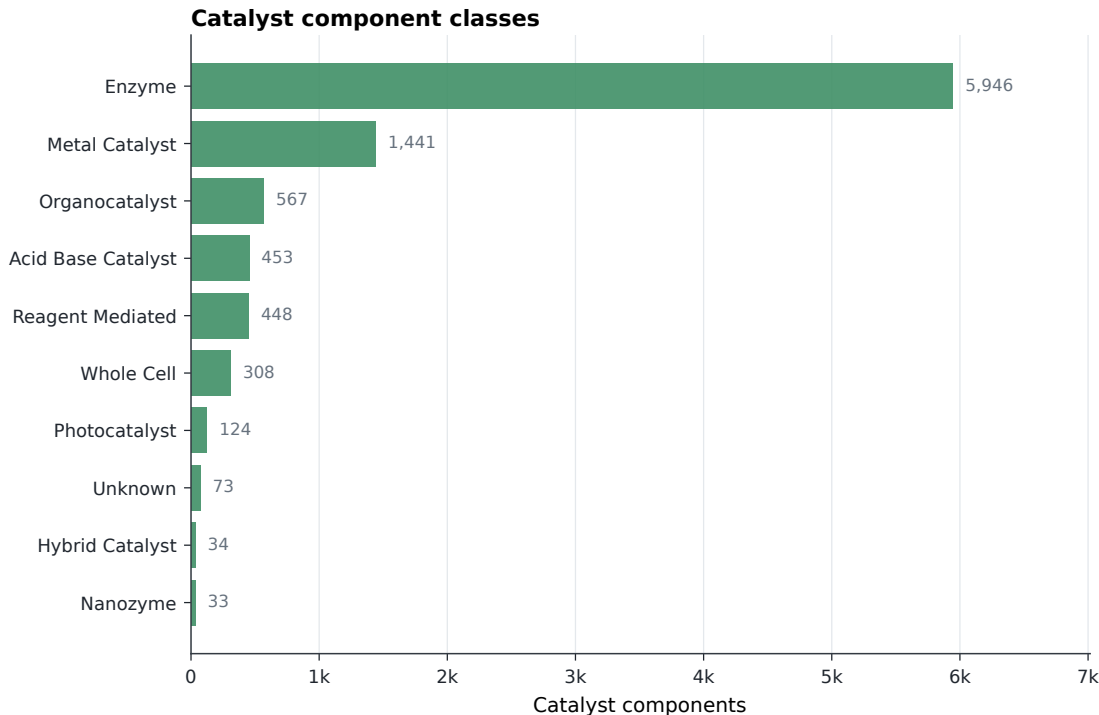


Figure 4: Catalyst component classes. Distribution of catalyst classes across normalized catalyst components.

The substrate-scope table contains 3,458 entries. Of these, 2,413 include yield percent, 672 include conversion percent, 2,151 include ee percent, and 2,989 include evidence or notes. These data enable users to separate single-condition record-level modelling from substrate-scope analyses that focus on matched families of substrates and outcomes.

5 Technical Validation

5.1 Schema and parsing checks

The quality audit found no JSON parse-error rows and no encoding-artifact flags in the v4 quality index. The release preserves both a high-quality subset and an all-quality audit index. This design supports conservative default reuse while retaining lower-tier records for retrieval, review, and future curation.

5.2 Field coverage

Top-level target product names are available for 3,806 of 3,810 source rows (99.90%), starting material names for 3,802 rows (99.79%), starting material SMILES for 3,396 rows (89.13%), target product SMILES for 3,051 rows (80.08%), overall yield for 2,301 rows (60.39%), overall ee for 1,769 rows (46.43%), and total reaction time for 2,983 rows (78.29%).

Condition coverage includes solvent name for 3,328 rows (87.35%), temperature for 2,803 rows (73.57%), pH for 1,681 rows (44.12%), and buffer name for 1,665 rows (43.70%). Step-level coverage

includes reaction SMILES for 6,759 of 8,753 parsed steps (77.22%), temperature for 7,079 steps (80.88%), reaction time for 5,694 steps (65.05%), pH for 4,312 steps, and step yield for 1,705 steps (19.48%). Catalyst identifier coverage includes EC numbers for 6,405 of 9,591 parsed catalyst components (66.78%) and UniProt identifiers for 4,174 components (43.52%).

5.3 Supplementary-information gains

The SI/LLM enrichment pass filled fields that are commonly absent from main-text extraction alone, especially reaction times, temperatures, pH values, substrate-scope entries, and quantitative outcomes. These gains are reported separately from baseline extraction so users can distinguish originally available fields from values added during targeted enrichment.

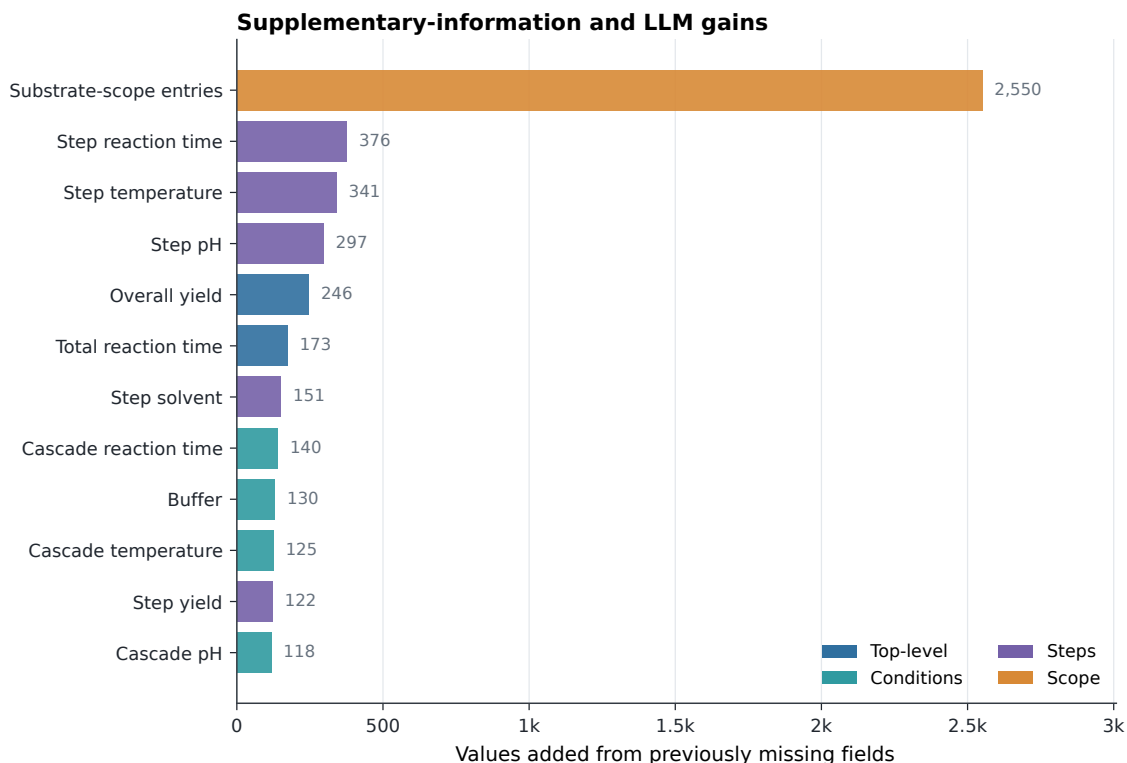


Figure 5: SI and LLM field gains. Values added from missing during the supplementary-information enrichment pass.

5.4 Backlog and residual risk

Supplementation backlog files separate remaining issues into 66 bronze or quarantine rows and 859 silver enhancement rows. The blocking backlog spans 51 unique DOI records; the silver enhancement backlog spans 673 unique DOI records. Among remaining backlogs, 35 blocking records and 551 enhancement records already have LLM OK status, indicating that further improvement may require stricter evidence alignment, manual review, or source-specific curation rather than simply running more extraction.

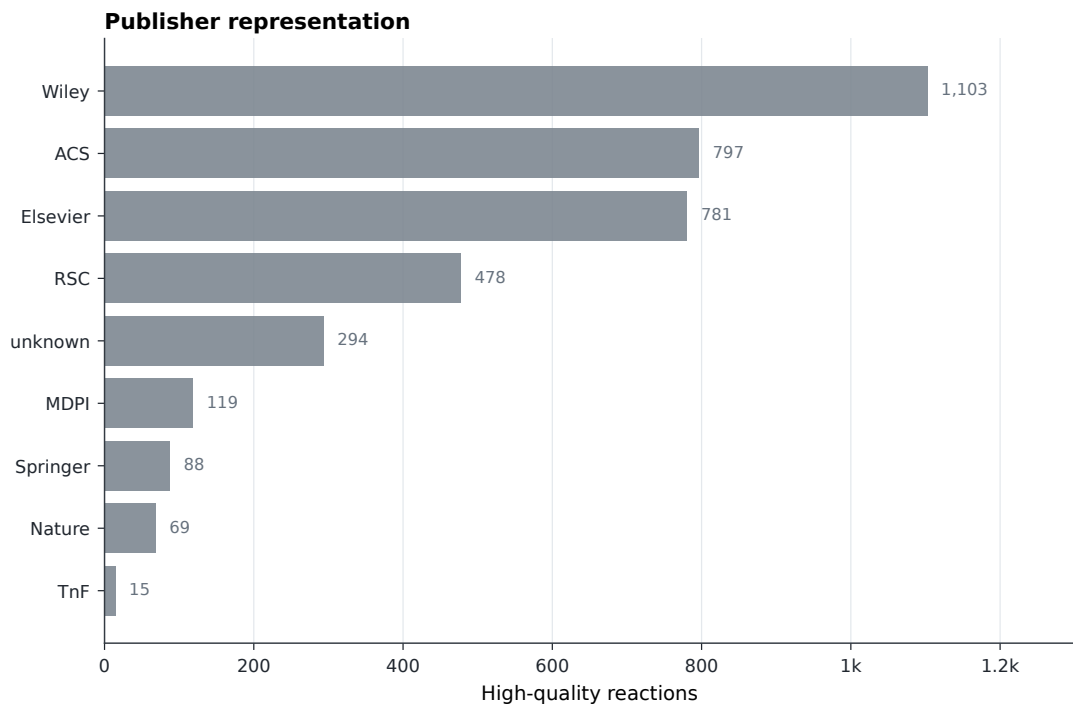


Figure 6: Publisher distribution. Publisher distribution for high-quality reaction records with article metadata.

5.5 Known limitations

Cascade reactions are heterogeneous across reporting styles, chemistry domains, and catalyst classes. Some silver records lack one or more enhancement dimensions such as quantitative outcomes, complete conditions, or SMILES. Step ordering and substrate-scope normalization can be ambiguous when source papers report variant conditions across dense supplementary tables. The dataset retains quality-tier labels, field-coverage outputs, and audit backlogs so that users can apply conservative filters for modelling or broader filters for retrieval and exploratory analysis.

6 Usage Notes

For training and modelling, users should start with the high-quality CSV, JSONL, or SQLite export. For strict model validation, users should restrict to the gold reaction table. For retrieval, literature review, or curation planning, users should use the all-quality index and backlog files. DOI-level provenance and quality-tier fields should be retained when creating downstream splits to avoid leakage across related records from the same source article.

Users interested in reaction-level modelling can use the flat high-quality CSV. Users interested in step-level transformation prediction, catalyst selection, or catalyst-class analyses should use the normalized **steps** and **catalysts** tables. Users interested in molecule-level modelling should require SMILES fields at the reaction, step, species, or substrate-scope level as appropriate. Users interested in process variables should inspect temperature, pH, solvent, buffer, and reaction-time coverage before fitting models.

7 Data Availability

The Cascade Dataset v4 release should be deposited in a persistent public repository before journal submission. The final repository DOI is DATA_REPOSITORY_DOI_TBD. If publisher full-text or supplementary content cannot be redistributed, the public repository should include derived metadata, extracted structured records, source DOI lists, audit outputs, and code sufficient to reproduce extraction for users with lawful access to the source articles.

8 Code Availability

The analysis code used to generate the manuscript assets will be archived with the data release or tagged in a public source-code repository. The final code repository URL or DOI is CODE_REPOSITORY_URL_OR_DOI_TBD.

9 Acknowledgements

[Acknowledgements to be added.]

10 Funding

[Funding information to be added.]

11 Author Contributions

[Author contribution statement to be added. Suggested structure: Conceptualization: X.X.; Data curation: X.X.; Software: X.X.; Validation: X.X.; Formal analysis: X.X.; Writing original draft: X.X.; Writing review and editing: all authors; Supervision: X.X.; Funding acquisition: X.X.]

12 Competing Interests

The authors declare no competing interests. [Confirm before submission.]

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A Release File Inventory

This appendix lists release files that should accompany the public data deposition. These names describe the archived data package rather than a local author workspace.

Resource group	File or table	Purpose
High-quality reaction export	<code>cascade_v4_high_quality.csv</code>	One row per gold or silver reaction for broad reuse.
Nested reaction export	<code>cascade_v4_high_quality.jsonl</code>	One JSON object per high-quality reaction, preserving nested steps, conditions, compatibility annotations, and substrate scope.
SQLite database	<code>cascade_v4_high_quality.db</code>	Normalized database containing reactions, steps, catalysts, species, and substrate-scope tables.
Tier-specific reactions	<code>cascade_v4_gold_reactions.csv</code> ; <code>cascade_v4_silver_reactions.csv</code>	Strict high-confidence subset and usable enhancement-gap subset.
Lower-tier records	<code>cascade_v4_bronze_reactions.csv</code> ; <code>cascade_v4_quarantine_reactions.csv</code>	Records retained for retrieval, audit, and future curation rather than default training.
Normalized tables	<code>cascade_v4_steps.csv</code> ; <code>cascade_v4_catalysts.csv</code> ; <code>cascade_v4_species.csv</code> ; <code>cascade_v4_substrate_scope.csv</code>	Step, catalyst, species, and substrate-scope tables derived from the high-quality subset.
All-quality audit index	<code>cascade_v4_all_quality_index.csv</code>	All 3,810 source rows with tier, flags, and provenance metrics.
Audit summaries	<code>audit_quality_summary.json</code> ; <code>audit_field_coverage_summary.csv</code> ; <code>audit_field_coverage_summary.json</code>	Quality-tier and field-coverage summaries.
Enrichment audit	<code>audit_detailed_field_increment.csv</code> ; <code>audit_detailed_field_increment.json</code>	SI/LLM field-gain summaries relative to the baseline database.
Backlog files	<code>audit_blocking_supplementation_backlog.csv</code> ; <code>audit_silver_enhancement_backlog.csv</code>	Remaining blocking and enhancement targets for future curation.
Documentation	<code>README.md</code> ; <code>DATA_DICTIONARY.md</code> ; <code>manifest.json</code>	Release overview, schema notes, generation metadata, file sizes, and checksums.

B Normalized Tables and Core Fields

reactions Top-level cascade metadata, DOI provenance, quality tier, source flags, target product, starting material, overall outcomes, global conditions, compatibility annotations, and record-level notes.

steps One row per cascade step, including step role, transformation superclass, transformation name, reaction SMILES, step conditions, and step outcome fields.

catalysts One row per catalyst component per step, including component name, catalyst class, EC number, UniProt identifier, organism or source annotations where available, and component-level provenance.

species Input and output species per step, including names, SMILES, roles, and linkage to the parent reaction and step.

substrate_scope Substrate-scope rows, including substrate/product labels, normalized outcome fields, evidence snippets or notes where available, and linkage to the parent reaction record.

C Quality Tiers and Recommended Filters

Gold Structurally complete cascade records with catalyst, substrate/product, conditions or outcome support, and no major quality flags. Use this tier for strict model validation.

Silver Usable cascade records with good structure but at least one enhancement gap, such as missing quantitative outcome, conditions, or SMILES. Combine gold and silver for the default high-quality release.

Bronze Retrieval- and curation-useful records that should not be used for default training unless manually reviewed.

Quarantine Structurally weak or suspicious records that should be excluded from training by default.

For reaction-level modelling, use the high-quality release and retain DOI and quality-tier fields for split construction. For strict molecular modelling, require the relevant substrate, product, species, or reaction SMILES fields. For process-property modelling, filter for temperature, pH, solvent, buffer, and reaction-time completeness. For catalyst selection, use the normalized catalyst table and apply EC-number or UniProt filters when sequence-level or enzyme-class information is required.